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JOURNAL OF COMBINATORIAL THEORY, Series A 37, 13–21 (1984)

Algebraic Constructions for Costas Arrays

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Communicated by the Managing Editors

Received May 27, 1982

1. INTRODUCTION

The following is equivalent to a problem posed by John P. Costas [1], who encountered it in the context of attempting to construct sonar signal patterns.

PROBLEM. For each positive integer n, construct an $n \times n$ permutation matrix with the property that the $\binom{n}{2}$ vectors connecting two 1's of the matrix are all distinct as vectors. (That is, no two vectors are equal in both magnitude and slope.)

Thus, if $a_{i_1,j_1} = a_{i_2,j_2} = a_{i_3,j_3} = a_{i_4,j_4} = 1$ in the matrix, we must not have $(i_2 - i_1,\ j_2 - j_1) = (i_4 - i_3, j_4 - j_3)$, nor may we have $(i_2 - i_1, j_2 - j_1) = (i_3 - i_2, j_3 - j_2)$.

Such matrices have been called either *Costas Arrays* or *constellations* in Ref. [2], which explores constructions as well as applications for these patterns. It is convenient to represent these arrays on an $n \times n$ grid, using dots for the 1's and blanks for the 0's of the matrix. Three examples of 6×6 Costas arrays are shown in Fig. 1.

Previous constructions [2] for Costas Arrays, for special values of n, have been discovered by L. R. Welch and by A. Lempel. This paper contains the

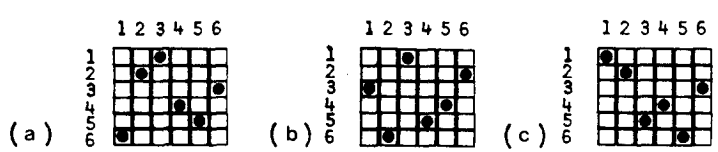


FIG. 1. Three examples of Costas Arrays of degree 6.

* This research was supported in part by the United States Navy, Office of Naval Research, under Contracts N00014-80-C-0745 and N00014-84-K-0189.

0097-3165/84 $3.00

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first published proofs of the validity of the Welch and Lempel constructions, as well as a major new construction.

As a consequence of all these constructions, $n \times n$ Costas Arrays are now known to exist for the following positive values of n:

(i) $n = p - 1, p$ prime,

(ii) $n = q - 2, q = p^k$ (any prime power),

(iii)* $n = q - 3, q = p^k$ (any prime power),

(iv)* $n = 2^k - 4, k \geqslant 3$.

Cases (iii) and (iv) depend on the conjecture that the field of q elements, $q > 2$, contains primitive roots α and β (not necessarily distinct) for which $\alpha + \beta = 1$. This conjecture has been verified for all prime powers q on the range $2 < q \leqslant 2^{11} = 2048$, and is discussed further in Section 3.

As an application of these results, algebraic constructions for $n \times n$ Costas Arrays are known for all $n \leqslant 130$ except for $n = 19, 31, 32, 33, 37, 43, 48, 49, 53, 54, 55, 63, 67, 73, 74, 75, 83, 84, 85, 89, 90, 91, 92, 93, 97, 103, 109, 113, 114, 115, 116, 117, 120, 121, 127$. However, the great majority of all integers n will not be included in these four classes of known constructions. It is hoped that this paper will inspire others to discover additional constructions for Costas Arrays.

2. The Basic Constructions

All known systematic constructions for Costas Arrays involve the use of primitive elements in finite fields.

THEOREM 1 (L. R. Welch). *Let g be a primitive root modulo the prime p. Then the $(p-1) \times (p-1)$ permutation matrix with $a_{ij} = 1$ iff $j \equiv g^i \pmod p$, $1 \leqslant i \leqslant p-1$, $1 \leqslant j \leqslant p-1$, is a Costas Array.*

Proof. Since g^i is a permutation of $\{1, 2, \ldots, p-1\}$ for $1 \leqslant i \leqslant p-1$, we have a permutation matrix. If it is not a Costas Array, we can find two pairs of points in it as follows:

$$\{(i, g^i), (i+k, g^{i+k})\}$$

$$\{(l, g^l), (l+k, g^{l+k})\}$$

where the corresponding vectors, $(k, g^{i+k} - g^i)$ and $(k, g^{l+k} - g^l)$ are equal, with $1 \leqslant k \leqslant p-2$. But this requires

$$g^i(g^k - 1) = g^l(g^k - 1),$$

and since $g^k - 1 \not\equiv 0 \pmod p$, we must have $i = l$. Hence the pattern is in fact a Costas Array. ∎

Note. This construction is illustrated in Fig. 1a with $p = 7$, $g = 3$.

LEMMA 1. *If an $n \times n$ Costas Array has a 1 in any of its four corners, the corresponding row and column can be removed to obtain an $(n-1) \times (n-1)$ Costas Array.*

Proof. Any violation of the Costas Array conditions in the reduced pattern would already be present in the original pattern. ∎

COROLLARY 1.1. *From the Welch Construction (Theorem 1), the left-most column and bottom row of the $(p-1) \times (p-1)$ matrix can be removed to obtain a $(p-2) \times (p-2)$ Costas Array.*

Proof. Since $g^{p-1} \equiv 1 (\mathrm{mod}\, p)$, the original array of degree $p-1$ has a 1 in position $(p-1, 1)$, i.e., at the lower left-hand corner. By Lemma 1, the array can then be reduced to degree $p-2$. ∎

COROLLARY 1.2. *If 2 is a primitive root modulo p, then a $(p-3) \times (p-3)$ Costas Array can be obtained from the Welch Construction.*

Proof. After removing the 1 at position $(p-1, 1)$, the 1 at position $(1, 2)$ is in the upper left-hand corner, and can also be removed by Lemma 1. ∎

COROLLARY 1.3. *Every cyclic permutation of the rows of a Costas Array in Theorem 1 is again a $(p-1) \times (p-1)$ Costas Array.*

Proof. Let g be a primitive root modulo p, and let c be any fixed positive integer. Then the $(p-1) \times (p-1)$ permutation matrix a_{ij} with $a_{ij} = 1$ if and only if $j = g^{i+c}$ is a Costas Array (the same proof as for Theorem 1), and the successive values of c give the successive cyclic permutations of the rows in the Theorem 1 construction. ∎

Note. It is clear that the Costas Array property is preserved under the group D_4 of dihedral symmetries of the $n \times n$ square.

THEOREM 2 (A. Lempel). *Let α be a primitive element in the field $GF(q)$, for any $q > 2$. Then the $(q-2) \times (q-2)$ symmetric permutation matrix with $a_{ij} = 1$ iff $\alpha^i + \alpha^j = 1$ is a Costas Array.*

Proof. It is quickly verified that this definition gives a symmetric permutation matrix of degree $q-2$. If $\alpha^i + \alpha^j = 1$, we may write $j = \log_\alpha(1 - \alpha^i)$. If the matrix is not a Costas Array, we can find two pairs of points in it as follows:

$$(i, \log_\alpha(1 - \alpha^i)), (i + k, \log_\alpha(1 - \alpha^{i+k}))\}$$

$$\{(l, \log_\alpha(1 - \alpha^l)), (l + k, \log_\alpha(1 - \alpha^{l+k}))\}$$

where the corresponding vectors,

$$\left(k, \log_\alpha \frac{1-\alpha^{i+k}}{1-\alpha^i}\right) \quad \text{and} \quad \left(k, \log_\alpha \frac{1-\alpha^{l+k}}{1-\alpha^l}\right),$$

are equal, with $1 \leqslant k \leqslant q-3$. But this requires each of the following:

$$\log_\alpha \left(\frac{1-\alpha^{i+k}}{1-\alpha^i}\right) = \log_\alpha \left(\frac{1-\alpha^{l+k}}{1-\alpha^l}\right),$$

$$\frac{1-\alpha^{i+k}}{1-\alpha^i} = \frac{1-\alpha^{l+k}}{1-\alpha^l},$$

$$1-\alpha^l-\alpha^{i+k}+\alpha^{i+l+k} = 1-\alpha^i-\alpha^{l+k}+\alpha^{i+l+k},$$

$$\alpha^{i+k}-\alpha^i = \alpha^{l+k}-\alpha^l$$

$$\alpha^i(\alpha^k-1) = \alpha^l(\alpha^k-1),$$

and since $\alpha^k - 1 \neq 0$, this requires $\alpha^i = \alpha^l$, and hence $i = l$. Hence the pattern is in fact a Costas Array. ∎

Note. This construction is illustrated in Fig. 1b, with $q = 8$, and α a root of $x^3 + x + 1 = 0$.

COROLLARY 2.1. *If 2 is a primitive root modulo p, then a symmetric* $(p-3) \times (p-3)$ *Costas Array can be obtained from the Lempel Construction of Theorem* 2.

Proof. Taking $q = p$ and $\alpha = 2$, we observe that $2^{p-1} \equiv 1 \pmod p$. Hence $2^{p-2} + 2^{p-2} \equiv 1 \pmod p$, so that $a_{p-2,p-2} = 1$. Since this is in the lower right-hand corner of the array, we invoke Lemma 1 to reduce the matrix to degree $p-3$.

THEOREM 3. *Let* α *and* β *be primitive elements in the field* $GF(q)$, *for any* $q > 2$. *Then the* $(q-2) \times (q-2)$ *permutation matrix with* $a_{ij} = 1$ *iff* $\alpha^i + \beta^j = 1$ *is a Costas Array. (This reduces to Theorem* 2 *in the special case* $\alpha = \beta$.)

Proof. From $\alpha^i + \beta^j = 1$ we may write $j = \log_\beta(1-\alpha^i)$. The rest of the proof then follows exactly (except for the base of the logarithms) the proof we have given for Theorem 2. ∎

THEOREM 4. *If the field* $GF(q)$ *contains two primitive roots,* α *and* β (*not necessarily distinct*) *with* $\alpha + \beta = 1$, *then the construction in Theorem* 3 *can be reduced to give a* $(q-3) \times (q-3)$ *Costas Array.*

Proof. Since $\alpha^1 + \beta^1 = 1$, we have $a_{11} = 1$ in the Costas Array obtained from Theorem 3. By Lemma 1, we may remove the top row and left column to obtain a Costas Array of degree $q - 3$. ∎

THEOREM 5. *If the field $GF(2^k)$ contains two primitive roots α and β with $\alpha + \beta = 1$, the Costas Array of degree $2^k - 2$ obtained from Theorem 3 can be successively reduced to obtain a Costas Array of degree $2^k - 3$ and a Costas Array of degree $2^k - 4$.*

Proof. From $\alpha^1 + \beta^1 = 1$, we square both sides to obtain $\alpha^2 + \beta^2 = 1$ (valid for fields of characteristic 2). Thus both $a_{11} = 1$ and $a_{22} = 1$. As in Theorem 4, we remove the top row and left column to obtain a Costas Array of degree $2^k - 3$. This new array has a 1 in the upper left-hand corner, and by Lemma 1 can be further reduced to degree $2^k - 4$. ∎

Note. Figure 1c illustrates Theorems 3, 4, and 5, using $q = 2^3 = 8$, with α a root of $x^3 + x + 1$ and $\beta = \alpha^3$. The Costas Array of degree 6 can be successively reduced to degrees 5 and 4 (and in fact, in this very special case, to degrees 3, 2, and 1 as well). In general, the Costas Arrays obtained by Theorems 3, 4, and 5 will be different as different choices of the primitive roots α and β are made.

CONJECTURE A. *It is conjectured that every finite field $GF(q)$ with $q > 2$ satisfies the condition of Theorem 4, namely, that it contains two primitive roots, α and β (not necessarily distinct) with $\alpha + \beta = 1$.*

We will examine the evidence for this conjecture in the next section.

3. A Conjectured Property of Finite Fields

Conjecture A, concerning all finite fields F with $q > 2$ elements, has the following equivalent formulations:

1. F contains primitive roots α and β (not necessarily distinct) with $\alpha + \beta = 1$.

2. F contains a primitive root α such that $1 - \alpha$ is also a primitive root.

3. Let $\Phi_{q-1}(x)$ be the cyclotomic polynomial of order $q - 1$, and degree $\phi(q - 1)$, over the rational field. Then considered as polynomials over F, $\Phi_{q-1}(x)$ and $\Phi_{q-1}(1 - x)$ have at least one common root in F.

A great deal is known (cf. [3–5]), about the factorization of cyclotomic polynomials over finite fields. The mapping $f(x) \to f(1 - x)$ preserves irreducibility, but not necessarily the primitivity of the roots.

If $q = 2^k$, $\Phi_{q-1}(x)$ factors over $F = GF(2)$ as a product of $\phi(2^k - 1)/k$ irreducible polynomials of degree k, all having roots which are primitive $(2^k - 1)$st roots of unity. Hence, $\Phi_{q-1}(1 - x)$ is also a product of $\phi(2^k - 1)/k$ irreducible polynomials of degree k. The *total* number of irreducible polynomials of degree k over $GF(2)$ is given by the "necklace number"

$$\frac{1}{k} \sum_{d|k} \mu(d)\, 2^{k/d}.$$

If the $\phi(2^k - 1)/k$ *primitive* irreducible polynomials constitute more than half of *all* the irreducible polynomials of degree k over $GF(2)$, then $\Phi_{2^k-1}(x)$ and $\Phi_{2^k-1}(1 - x)$ must have some (at least one) of these polynomial factors in common modulo 2, and hence some (at least k) common roots in $GF(2^k)$. This is a *sufficient* condition for Conjecture A to hold for $GF(2^k)$. The first (smallest) such field for which this sufficient condition does not obtain is $GF(2^{12})$, where about 43% of the irreducible polynomials are primitive. Even in this case, it is overwhelmingly likely that many of the 144 primitive polynomials of degree 12 are mapped into other primitive polynomials of degree 12 by the mapping $x \to 1 - x$. As a simple counting argument, we have:

THEOREM 6. *If more than half the elements of $GF(q)$ which are in no proper subfield of $GF(q)$ are primitive $(q - 1)$st roots of unity, then Conjecture* A *holds in $GF(q)$. If $GF(q)$ has no proper subfields, the elements* 0 *and* 1 *may be excluded.*

Proof. $x \to 1 - x$ is a permutation on the elements of $GF(q)$ which interchanges 0 and 1, and leaves proper subfields setwise fixed. If more than half the remaining elements are primitive, $x \to 1 - x$ must take at least one primitive element into another. ∎

Note. Among fields $GF(q)$ which are instances of this theorem are (i) those with $q = 2^k$ where more than half the irreducible polynomials of degree k over $GF(2)$ have roots which are primitive $(2^k - 1)$st roots of unity; (ii) $GF(F_n)$ where $F_n = 2^{2^n} + 1$ is a Fermat prime; and (iii) $GF(9)$.

For general q, there are $\phi(q - 1)$ primitive roots in the field of q elements, and $\binom{\phi(q-1)+1}{2}$ pairs of primitive roots, α and β (not necessarily distinct). Conjecture A requires that only one of these pairs satisfy $\alpha + \beta = 1$. By a statistical randomness argument, it would seem extremely unlikely that Conjecture A could fail for very large q. Under the circumstances, it may be significant that Conjecture A has been verified for all *small* q $(2 < q \leqslant 2^{11})$. The largest value of q on this range for which the pair (α, β) of primitive roots with $\alpha + \beta = 1$ was unique was $q = 43$.

THEOREM 7 (H. Taylor). *If p and r are odd primes such that $p = 2^m \cdot r + 1$, and $r > 2^{m-1} + 2$, then p has primitive roots α and β such that $\alpha + \beta = 1$ in $GF(p)$.*

Proof. Case $m = 1$. In this case $p \equiv -1 \pmod 4$ and whenever x is a primitive root, it necessarily follows that $-x$ is not a primitive root.

Arranging the non-zero elements of $GF(p)$ as shown below, we are representing them with the integers from $-r$ to r.

$$\begin{array}{cccccccc} 1 & 2 & 3 & \cdot & \cdot & \cdot & \cdots & r \\ -1 & -2 & -3 & \cdot & \cdot & \cdot & \cdots & -r \end{array}$$

We will find *at most one* primitive root in each column, and of course none in the first column.

Since the number of primitive roots is $\phi(2r) = r - 1$, there must in fact be exactly one primitive root in each column except the first.

If $-r$ is a primitive root we take $\alpha = \beta = -r$, and have $\alpha + \beta = 1$. Otherwise r is a primitive root and we look for a primitive root in the "minus" row. Indeed, -4 is always a primitive root because 4 is a quadratic residue, and $r > 3$. There must be some k, with $4 \leqslant k < r$, such that $-k$ is the primitive root in the kth column, and $k + 1$ is the primitive root in the $(k + 1)$st column. At that point we can take $\alpha = -k$ and $\beta = k + 1$.

Proof. Case $m \geqslant 2$. In this case $p \equiv 1 \pmod 4$, and whenever x is a primitive root, it necessarily follows that $-x$ is also a primitive root. Thus in each column shown below we will find that both entries are primitive roots, or else neither is:

$$\begin{array}{ccccccccc} 1 & 2 & 3 & 4 & 5 & \cdot & \cdot & \cdot & 2^{m-1} \cdot r \\ -1 & -2 & -3 & -4 & -5 & \cdot & \cdot & \cdot & -2^{m-1} \cdot r \end{array}$$

If $2^{m-1} \cdot r$ is a primitive root we take $\alpha = \beta = -2^{m-1} \cdot r$, and have $\alpha + \beta = 1$ as desired.

If $2^{m-1} \cdot r$ is not primitive, we look for k such that column k has both entries primitive and column $k + 1$ also has both entries primitive. When this happens we can take $\alpha = -k$ and $\beta = k + 1$. To show that this *must* happen, we use a form of the pigeonhole principle, as follows: Strictly between an odd numbered column and an even numbered column, the number of columns will be even. An even number of consecutive columns can hold at most as many primitive roots as there are columns, if it must avoid two consecutive occupied columns.

Since quadratic residues cannot be primitive, we know that columns 1^2, 2^2, 3^2,..., $(2^{m-1} + 1)^2$ cannot hold primitive roots. Also these column numbers alternate odd-even ending with odd, so that together with the last

column (even-numbered $2^{m-1} \cdot r$), they leave room for no more than $2^{m-1} \cdot r - 2^{m-1} - 2$ primitive roots.

Since $r \geqslant 2^{m-1} + 2 + 1$ by hypothesis, we know that $2^{m-1} \cdot r \geqslant (2^{m-1})^2 + 2^m + 2^{m-1} > (2^{m-1} + 1)^2$, so that the last column is safely beyond the $(2^{m-1} + 1)^2$ column.

But the total number of primitive roots is $\phi(2^m \cdot r) = 2^{m-1} \cdot r - 2^{m-1}$, which is larger than the number of primitive roots which the columns can hold without having two consecutive columns occupied. Thus there must exist primitive roots α and β such that $\alpha + \beta = 1$ in $GF(p)$. ∎

Analogous to Conjecture A, we also have:

CONJECTURE B. *Every finite field $GF(q)$ with $q > 3$ contains two primitive roots α and β with $\alpha + \beta = -1$.*

Note. Conjecture B has also been verified for all q on the range $2 < q \leqslant 2^{11}$. Of course, Conjecture B reduces to Conjecture A if $q = 2^k$, where $+1 = -1$; and also if $q \equiv 1 \pmod 4$, since in this case α and β are primitive iff $-\alpha$ and $-\beta$ are also primitive, so that $\alpha + \beta = 1$ is an instance of Conjecture A iff $(-\alpha) + (-\beta) = -1$ is an instance of Conjecture B.

For large q, it is reasonable to expect the following:

CONJECTURE C. *For all $q > q_0$, every non-zero element ε of $GF(q)$ has at least one representation of the form $\varepsilon = \alpha + \beta$, where α and β are primitive elements of $GF(q)$.*

Note. $\alpha + \beta = 0$ will *not* occur if $q \equiv -1 \pmod 4$.

4. THE CASE OF CONJUGATE QUADRATIC ROOTS

In Theorem 3, we saw that the solutions (i, j) of $\alpha^i + \beta^j = 1$ generate a $(q-2) \times (q-2)$ Costas Array whenever α and β are primitive roots in $GF(q)$. If $q = p^{2m}$, then $GF(q)$ may be regarded as a quadratic extension of $GF(p^m)$. In this case, there will always be at least one irreducible quadratic equation $x^2 + ax + b$ over $GF(p^m)$ whose roots α and β are primitive elements in $GF(q)$. In this case, it is known that $\alpha = \beta^{p^m}$ and $\beta = \alpha^{p^m}$. Hence we have the following result.

THEOREM 8. *Suppose α and β are primitive elements of $GF(p^{2m})$ which are conjugate roots of the same irreducible quadratic over $GF(p^m)$. Then the solutions (i, j) of $\alpha^i + \beta^j = 1$ generate a symmetric $(q-2) \times (q-2)$ Costas Array, where $q = p^{2m}$.*

Proof. Theorem 3 yields everything except the symmetry of the array.

We must show that whenever $a_{ij} = 1$ then also $a_{ji} = 1$. Now $a_{ij} = 1$ iff $\alpha^i + \beta^j = 1$. We raise both sides of this equation to the power p^m. Thus $(\alpha^i + \beta^j)^{p^m} = 1$, $(\alpha^i)^{p^m} + (\beta^j)^{p^m} = 1$, $(\alpha^{p^m})^i + (\beta^{p^m})^j = 1$, $\beta^i + \alpha^j = 1$. That is, whenever $a_{ij} = 1$, then also $a_{ji} = 1$. ∎

If we can further impose the requirement that $\alpha + \beta = 1$, then the symmetric $(q-2) \times (q-2)$ array can be reduced to a symmetric $(q-3) \times (q-3)$ array. (In the case of characteristic 2, this can be further reduced to a $(q-4) \times (q-4)$ array.) Based on verification for all $q < 2400$, we have:

CONJECTURE D. *There exists a primitive quadratic of trace* 1 *over every finite field* $GF(q)$.

That is, there is an irreducible quadratic polynomial $x^2 - x + g$ over $GF(q)$ whose roots α and β are primitive elements in the multiplicative group $GF(q^2)$.

Note that Conjecture D implies Conjecture A whenever the q of Conjecture A is an *even* power of a prime. Although this makes Conjecture D stronger than Conjecture A for the fields $GF(p^{2m})$, it may be easier to prove Conjecure D. For example, the following similar result has been proved (see [6]): "There is a primitive polynomial of trace 1 for every degree $n > 1$, over $GF(2)$." This result was first proved using the work of Davenport [7].

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